



COMPUTER VISION IN HEALTHCARE

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VISION

INTRODUCTION

- Branch Of AI That Enables Computers To Understand Medical Images.
- Assists Doctors In Diagnosis.
- Improves Accuracy And Speed.
- Reduces Human Error.
- Used In Hospitals And Research.



Tools Used in Computer Vision for Healthcare

OpenCV: Image preprocessing and feature extraction.

TensorFlow: Develops AI models for medical image classification.

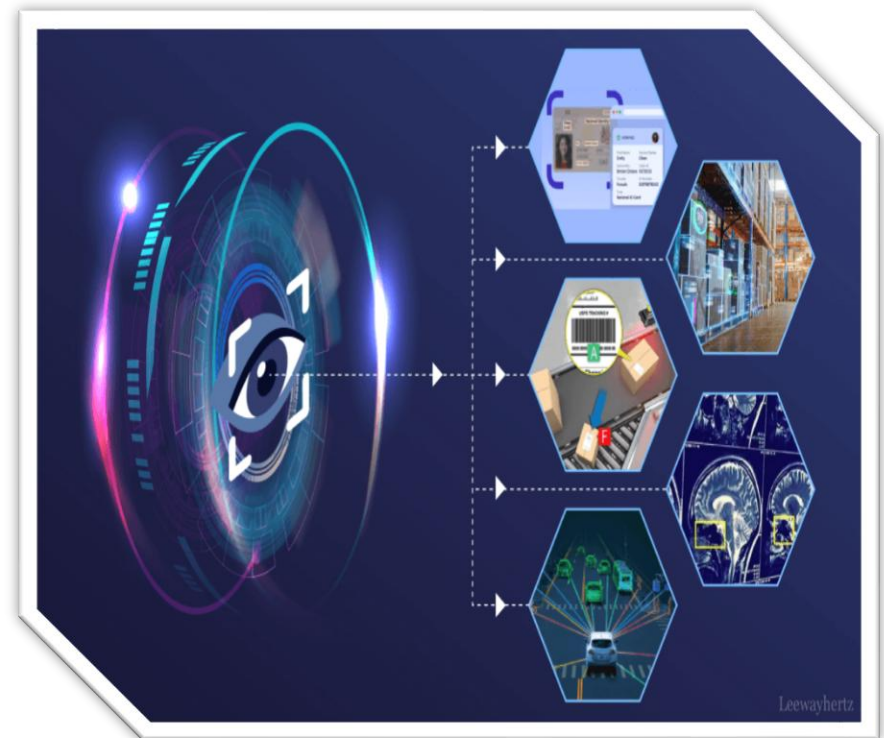
PyTorch: Builds deep learning models for image analysis.

CNN: Detects and classifies diseases from medical images.

DICOM: Stores and transfers medical imaging data.

HOW COMPUTER VISION WORKS IN HEALTHCARE

- 1. Image Acquisition (X-ray, MRI, CT).
- 2. Image Preprocessing.
- 3. Feature Extraction.
- 4. AI Disease Detection.
- 5. Diagnosis Report.



HOW COMPUTER VISION IN HEALTHCARE WORKS STEP-BY-STEP

Image
acquisition

Preprocessing

Feature
extraction

Feature
representation

Recognition
and
interpretation

Post-processing
and decision-
making

APPLICATIONS



- **Cancer Detection:** Identifies tumors and abnormal cells from medical images for early diagnosis.
- **Brain Tumor Detection:** Detects and locates brain tumors using MRI and CT scans.
- **Diabetic Retinopathy Detection:** Examines retinal images to identify diabetes-related eye diseases.
- **Fracture Detection:** Detects bone fractures quickly and accurately from X-ray images.
- **Surgical Assistance:** Provides real-time image guidance to improve surgical precision.
- **Patient Monitoring:** Tracks patient movements and vital conditions for continuous care.
- **Telemedicine:** Enables remote diagnosis and consultation through AI-based medical image analysis.

ADVANTAGES

- **Faster Diagnosis:** Detects diseases quickly using AI.
- **High Accuracy:** Provides accurate analysis of medical images.
- **Early Disease Detection:** Identifies diseases at an early stage.
- **Reduces Human Error:** Minimizes mistakes in diagnosis.
- **Saves Doctors' Time:** Automates image analysis for faster decisions.
- **Better Patient Care:** Improves treatment quality and patient outcomes.

CHALLENGES

- **Requires Large Datasets:** Needs a large amount of medical data for accurate training.
- **High Implementation Cost:** Requires expensive software, hardware, and maintenance.
- **Privacy Concerns:** Patient medical data must be protected from unauthorized access.
- **Needs Powerful Hardware:** Requires high-performance computers and GPUs for processing.
- **Depends on Image Quality:** Poor-quality images can reduce detection accuracy.
- **Ethical Issues:** AI decisions must be fair, transparent, and used responsibly.

CONCLUSION

- Improves healthcare through AI.
- Supports doctors, not replaces them.
- Enhances diagnosis and treatment.
- Future of smart healthcare.